

Triggers for Stock Management in a Shopping Center

CODE:

```
Create database TRI;
Use TRI;
CREATE TABLE Product (
ProductID INT PRIMARY KEY,
ProductName VARCHAR(100) NOT NULL,
Stock INT NOT NULL CHECK (Stock >= 0)
);
CREATE TABLE Coupon (
CouponID INT PRIMARY KEY,
CouponCode VARCHAR(50) UNIQUE NOT NULL
);
CREATE TABLE Sales (
SaleID INT AUTO_INCREMENT PRIMARY KEY,
ProductID INT NOT NULL,
Qty INT NOT NULL CHECK (Qty > 0),
CouponID INT,
FOREIGN KEY (ProductID) REFERENCES Product(ProductID),
FOREIGN KEY (CouponID) REFERENCES Coupon(CouponID)
);
INSERT INTO Product (ProductID, ProductName, Stock) VALUES
(1, 'Laptop', 50),
(2, 'Mouse', 200),
(3, 'Keyboard', 30);
INSERT INTO Coupon (CouponID, CouponCode) VALUES
(1, 'DISCOUNT10'),
(2, 'FREESHIP'),
(3, 'SUMMER50');
DELIMITER $$
CREATE TRIGGER BeforeDeleteProduct
BEFORE DELETE ON Product
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FOR EACH ROW
BEGIN
IF OLD.Stock > 0 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT = 'Cannot delete product with stock greater than 0.';
END IF;
END $$
DELIMITER ;
DELIMITER $$
CREATE TRIGGER AfterInsertSale
AFTER INSERT ON Sales
FOR EACH ROW
BEGIN
-- Delete the used coupon code
IF NEW.CouponID IS NOT NULL THEN
DELETE FROM Coupon WHERE CouponID = NEW.CouponID;
END IF;
-- Use a session variable to store the current stock
SET @currentStock = (SELECT Stock FROM Product WHERE ProductID =
NEW.ProductID);
-- Check if the product stock allows the sale
IF (@currentStock - NEW.Qty) < 0 THEN
SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT = 'Cannot sell more products than available in stock.';
END IF;
-- Update the product stock
UPDATE Product
SET Stock = Stock - NEW.Qty
WHERE ProductID = NEW.ProductID;
END $$
DELIMITER ;
-- Attempt to delete a product with stock
DELETE FROM Product WHERE ProductID = 1;
-- Error: "Cannot delete product with stock greater than 0."
-- Reduce stock to 0 and try again

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UPDATE Product SET Stock = 0 WHERE ProductID = 1;
DELETE FROM Product WHERE ProductID = 1; -- Success
select * from Product;
-- Attempt to add a sale exceeding stock
INSERT INTO Sales (ProductID, Qty, CouponID) VALUES (2, 300, 1);
-- Error: "Cannot sell more products than available in stock."
-- Add a valid sale
INSERT INTO Sales (ProductID, Qty, CouponID) VALUES (2, 50, 2);
-- Coupon 2 is removed, and stock of ProductID 2 is updated to 150.
ALTER TABLE Sales DROP FOREIGN KEY sales_ibfk_2;
ALTER TABLE Sales ADD CONSTRAINT sales_ibfk_2 FOREIGN KEY
(CouponID) REFERENCES Coupon(CouponID) ON DELETE CASCADE;
DELETE FROM Product WHERE ProductID = 1; -- This will fail if stock > 0
UPDATE Product SET Stock = 0 WHERE ProductID = 1; -- Reduce stock to 0
DELETE FROM Product WHERE ProductID = 1; -- This should succeed now

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SELECT * FROM Product;
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INSERT INTO Sales (ProductID, Qty, CouponID) VALUES (2, 300, 1);
INSERT INTO Sales (ProductID, Qty, CouponID) VALUES (2, 50, 2);

```

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SELECT * FROM Coupon;
select * from Sales;

```

OUTPUT:

✓	37	14:09:19	DELETE FROM Product WHERE ProductID = 1	0 row(s) affected	0.000 sec
✓	38	14:09:22	SELECT * FROM Product LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
✗	39	14:09:28	INSERT INTO Sales (ProductID, Qty, CouponID) VALUES (2, 300, 1)	Error Code: 1644. Cannot sell more products than available in stock.	0.110 sec
✓	40	14:09:33	INSERT INTO Sales (ProductID, Qty, CouponID) VALUES (2, 50, 2)	1 row(s) affected	0.015 sec
✓	41	14:09:37	SELECT * FROM Coupon LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec
✓	42	14:10:07	select * from Sales LIMIT 0, 1000	0 row(s) returned	0.000 sec / 0.000 sec
✗	43	14:10:38	INSERT INTO Sales (ProductID, Qty, CouponID) VALUES (2, 300, 1)	Error Code: 1644. Cannot sell more products than available in stock.	0.016 sec
✓	44	14:11:10	SELECT * FROM Product LIMIT 0, 1000	2 row(s) returned	0.000 sec / 0.000 sec

ProductID	ProductName	Stock
2	Mouse	150
3	Keyboard	30
NULL	NULL	NULL

	CouponID	CouponCode
▶	1	DISCOUNT10
	3	SUMMER50
✱	NULL	NULL

SaleID	ProductID	Qty	CouponID
1	2	50	2